

Jinglin ZHANG

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EDUCATION

Duke University, Duke Kunshan University 2021 – 2023

Master of Environmental Policy

Durham, United States; Suzhou, China.

Aberystwyth University 2018 – 2020

Bachelor of Science in Environmental Science with Second Class Honors, Division 1

Aberystwyth, United Kingdom.

Hohai University 2016 – 2018

Bachelor of Engineering in Environmental Science

Nanjing, China

ACADEMIC EXPERIENCE

Research Fellow in Environmental Economics 2023 – 2026

Duke Kunshan University

Report to: Prof. Jingbo Cui

- Corporate exposure to climate risks and the synergy between conservation and economic development.
- Data collection, cleaning, analysis, visualization, report drafting, assist funding application and administration.

PROJECT: Would Extreme Weather Backfire Corporate Carbon Mitigation Efforts: Evidence from Chinese Firm-level Data (in progress)

supervisor: Prof. Jingbo Cui

- Calculate annual carbon emissions of Chinese companies from 2008 to 2016 and identify annual rainfall and temperature distributions.
- Using fixed effects panel regression models to reveal the companies response to extreme heat and drought in carbon emissions, identify the heterogeneity of responses across different sectors and sizes, and predict future carbon emissions trends for companies under different CMIP6 scenarios.
- First empirical evidence of environmental impacts of extreme weather on carbon emission in developing countries.

PROJECT: Spatiotemporal Dynamics and Driving Factors of *Larix chinensis* Plantations in the Qinling Region (in progress)

supervisor: Prof. Binbin V. Li

- The follow-up research for the master's thesis.
- Classification of larch plantation based on multi-temporal Sentinel-2 images in Southern Qinling Region, compare the performance for different classification algorithms: random forest, supervised / self-supervised dual LSTM.
- Identity the distribution of larch patches when planting using multi-temporal Landsat-5 images from 1980 to 2000, compare the performance for different land use change algorithms: CCDC, LandTrendr, VCT.
- Calculate landscape indexes for current and historical larch plantations, use OLS to identify factors driving landscape change.
- First to analyze the drivers of plantation change in China from a landscape dynamics perspective.

PROJECT: Distribution Law and Transport Mechanism of Pollutants in Streams in the Plain River Network Area of Taihu Lake Basin

Jiangsu Students' Innovation Training Program (Provincial Excellent Program)

- Work with team to do field observation, measured the section area, flow velocity and water level of the rivers and constructed a two-dimensional hydrodynamic model of the research area with EFDC Explorer.
- Investigating influencing factors and the mechanism of the flow diversion ratio in the plain river network.
- Joined publish the paper: Wang, Z., Li, Y., Weng, S., **Zhang, J.**, & Hu, T. (2019, November). Research on the Water Diversion Ratio Characteristics of the Plain River Network--A Case Study of the Channel in Zhangjiagang. In *IOP Conference Series: Earth and Environmental Science* (Vol. 384, No. 1, p. 012190). IOP Publishing.

UNDERGRADUATE DISSERTATION: The Influence of Two Biofuel Crops (miscanthus and salix spp.) on Soil Biodiversity

supervisor: Prof. Peter Dennis

- Collect and identify the biodiversity of invertebrates in soil samples from different crops and planting patterns, as well as the physical and chemical properties of the soil in the samples
- Compare soil biodiversity and soil physicochemical properties of two crops under concentrated and mixed cropping patterns. The result reveals the soil biodiversity was higher under mixed planted pattern.

FIELDWORK: Biodiversity-friendly tea plantations

Yunnan, China | Sept. 2024

- Assist in collecting biodiversity data with bird point counts, plant quadrat surveys, and camera traps.

FIELDWORK: population survey on *Onychoprion fuscatus* in Dry Tortugas National Park

Florida, USA | Mar. 2023

- Captured and banded sooty terns in Dry Tortugas National Park.

FIELDWORK: Distribution and Biodiversity Survey of Larch Plantations in the Qinling Mountains

Shaanxi, China | July 2022 – Aug. 2022

- Visited ten protected areas and forest farms in the southern Qinling Mountains, investigated the distribution of larch plantation in these places, established sites suitable for biodiversity survey, and conducted quadrat surveys for four to five weeks, and measured the and secondary forest vegetation biodiversity.

FIELDWORK: Biodiversity survey in Wanglang National Nature Reserve

Sichuan, China | June 2022

- Conducted quadrat surveys in Wanglang Nature Reserve, collected vegetation biodiversity data, collected infrared cameras, interpreted infrared camera data, participated in social surveys, and collected opinions of indigenous community on grazing policies.

FIELDWORK: ES20320 Environmental Science Fieldwork Course

Republic of Ireland | Apr. 2019

- Assessed biomass and woodland ecology of the park, geomorphic mapping of local woodland, fluvial hydrology, water quality and macroinvertebrates biodiversity of local river.

TEACHING EXPERIENCE

Instructor for GIS Workshop

Aug. 2024

iMEP Summer School 2024

- Taught fundamental GIS concepts and applications, spatial data visualization and map design; guided students in using ArcGIS Pro to perform basic multi-criteria analysis (MCA) exercises.

NON-ACADEMIC EXPERIENCE

Public Service Track, Student Worker <i>Duke Kunshan University Library</i> • Provided supports to users who use library equipment and system. • Shelved and maintained printed collection, assisted in library activities.	Apr. 2022 – Dec. 2022
Academic Representative of F750 Environmental Science <i>Aberystwyth University</i> • Took the lead on collecting and voicing student feedback to staff. • Work with faculty and Academic Affair Officers to raise feedbacks and suggestions to enhance the service and experience in the university.	Sept. 2018 – June 2020

SCHOLARSHIP

Guo Tingting Scholarship CNY 10,000	2022
International Master of Environmental Policy Scholarship CNY 54,000 in total	2021-2023
Aberystwyth Regional Scholarships for Undergraduate Students GBP 2,000 in total	2019-2020

ADDITIONAL INFORMATION

LANGUAGES

Chinese: Native
English: IELTS 6.5 (2018).

SOFTWARE

- General: Python, R, Microsoft Office.
- GIS: ArcGIS Pro; Google Earth Engine, ENVI.
- Data Analysis: STATA, SPSS.
- Hydrology: EFDC Explorer.